



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2013

**TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION**

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes.
- Working time – 180 minutes.
- Write using black or blue pen.
- Board approved calculators may be used.
- Show all necessary working in

Questions 11–16

Total Marks - 100 Marks

Section I 10 Marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section.

Section II 90 Marks

- Attempt Questions 11–16
- Allow about 2 hour 45 minutes for this section.

Examiner: *External Examiner*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1) Given $z = r(\cos \theta + i \sin \theta)$, then $\left| \frac{\bar{z}}{z^2} \right|$ equals
- (A) r
- (B) r^2
- (C) $\frac{1}{r}$
- (D) $-r$
- 2) If $1 + i$ is a root of the polynomial $x^3 - 4x^2 + 6x - 4 = 0$. The other roots are:
- (A) $1 - i$ and -2
- (B) $1 + i$ and -2
- (C) $1 - i$ and 2
- (D) $1 + i$ and 2
- 3) If the polynomial equation $P(x) = 0$, has roots α, β, γ then the roots of the polynomial equation $P(3x + 2) = 0$ are
- (A) $\frac{\alpha}{3} - 2, \frac{\beta}{3} - 2, \frac{\gamma}{3} - 2$
- (B) $\frac{\alpha-2}{3}, \frac{\beta-2}{3}, \frac{\gamma-2}{3}$
- (C) $3\alpha + 2, 3\beta + 2, 3\gamma + 2$
- (D) $\alpha + \frac{2}{3}, \beta + \frac{2}{3}, \gamma + \frac{2}{3}$
- 4) The gradient of the tangent to the curve $2x^3 - y^2 = 7$ at the point $(2, -3)$ is:
- (A) -4
- (B) -2
- (C) 2
- (D) 4

- 5) The area bounded by the parabola $x^2 = 4ay$ and the line $y = a$ is rotated about the line $y = a$. To find the volume of the resulting solid, the slicing technique is used.

The area of a typical slice is given by

- (A) $\pi(a - y)^2$
(B) $\pi(a^2 + y^2)$
(C) $\pi(a - x)^2$
(D) $\pi(a^2 + x^2)$
- 6) The equation of the conic whose distance from the point $(1,0)$ is half its distance from the line $x = 4$ is given by:

- (A) $\frac{x^2}{4} + \frac{y^2}{3} = 1$
(B) $\frac{x^2}{4} + \frac{y^2}{4} = 1$
(C) $\frac{x^2}{4} - \frac{y^2}{3} = 1$
(D) $\frac{x^2}{3} - \frac{y^2}{4} = 1$

- 7) The number of different arrangements of the letters of the word SERVICES which begin and end with letter S is:

- (A) $\frac{8!}{2!}$
(B) $\frac{6!}{2!}$
(C) $\frac{6!}{(2!)^2}$
(D) $\frac{8!}{(2!)^2}$

8) Given the curve $y = f(x)$, then the curve $y = f(|x|)$ is represented by

- (A) A reflection of $y = f(x)$ in the y -axis
- (B) A reflection of $y = f(x)$ in the x -axis
- (C) A reflection of $y = f(x)$ for $x \geq 0$ in the y -axis
- (D) A reflection of $y = f(x)$ for $y \geq 0$ in the x -axis

9) Using the substitution $x = \pi - y$, the definite integral

$$\int_0^{\pi} x \sin x \cdot dx$$

will simplify to:

- (A) 0
- (B) $\frac{\pi^2}{4}$
- (C) $\frac{\pi}{2} \int_0^{\pi} \sin x \cdot dx$
- (D) $\int_0^{\pi} \sin x \cdot dx$

10) Which of the following statements is false?

- (A) $\int_{-3}^3 x^3 e^{-x^2} \cdot dx = 0$
- (B) $\int_{-4}^4 \frac{x^2}{x^2+4} \cdot dx = 2 \int_0^4 \frac{x^2}{x^2+4} \cdot dx$
- (C) $\int_0^{\pi} \sin^4 \theta \cdot d\theta > \int_0^{\pi} \sin 4\theta \cdot d\theta$
- (D) $\int_0^1 x^4 \cdot dx < \int_0^1 x^5 \cdot dx$

End of Section I

Section II

Free response questions

Total marks – 90

Attempt Questions 11 – 16

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 11 (15 marks) Use a SEPARATE writing booklet.

- (a) Find

$$\int x \tan^{-1} x \cdot dx \quad 2$$

- (b) Use completion of squares to evaluate

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{dx}{\sqrt{15 - 4x - 4x^2}} \quad 2$$

- (c) Use the substitution $t = \tan \frac{\theta}{2}$ to evaluate

$$\int_{\frac{\pi}{2}}^{\frac{2\pi}{3}} \frac{d\theta}{2 \cot \frac{\theta}{2} - \sin \theta} \quad 2$$

- (d)

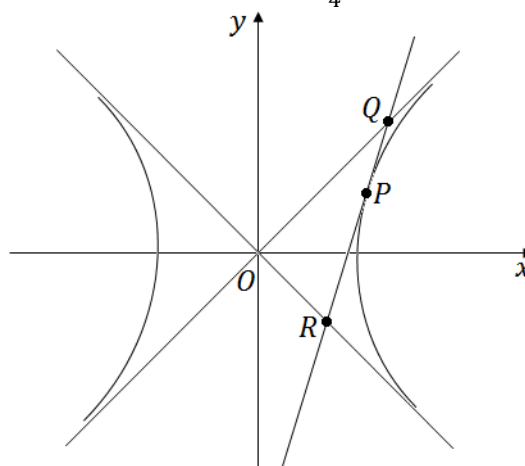
- (i) Find the real number A, B and C such that

$$\frac{3x^2 - x + 8}{(1 - x)(x^2 + 1)} = \frac{A}{1 - x} + \frac{Bx + C}{x^2 + 1} \quad 2$$

- (ii) Hence find

$$\int \frac{3x^2 - x + 8}{(1 - x)(x^2 + 1)} \cdot dx \quad 2$$

- (e) $P(2 \sec \theta, \tan \theta)$ is a point on the hyperbola $H: \frac{x^2}{4} - y^2 = 1$



If the tangent at P cuts the asymptotes at Q and R as shown in the figure above, find the coordinates of Q and R in terms of θ . Show that P is the mid-point of QR . 3

- (f) Find a and b where a and b are real numbers if $(a + ib)^2 = 21 - 20i$. 2

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) Given that α, β and γ are the roots of the equation

$$2x^3 + 3x^2 - 5x + 8 = 0 \quad 2$$

find the polynomial equation with roots $\frac{1}{\alpha}, \frac{1}{\beta}$ and $\frac{1}{\gamma}$.

- (b) Let $z = 3(\cos \theta + i \sin \theta)$

(i) Find $\overline{1 - z}$ 1

(ii) Express the imaginary part of $\frac{1}{1-z}$ in terms of θ . 1

- (c) Sketch the region for z in the Argand plane defined by:

$$|z - 1 + i| < 2 \quad \text{and} \quad -\frac{\pi}{4} \leq \arg(z - 1 + i) \leq \frac{5\pi}{4} \quad 2$$

- (d) If $z_1 = 2i$ and $z_2 = 1 + 3i$ are two complex numbers, describe the loci of z such that:

$$z = z_1 + k(z_2 - z_1), \text{ when}$$

(i) $k = 1$ 1

(ii) $0 < k < 1$ 1

(iii) k is any real number. 1

- (e) The polynomial $P(x) = x^3 + ax + b$ has zeroes α, β and $2(\alpha - \beta)$.

(i) Show that $a = -13\alpha^2$ and $b = 12\alpha^3$. 2

(ii) Deduce that the zeroes of $P(x)$ are $-\frac{13b}{12a}, -\frac{13b}{4a}$ and $\frac{13b}{3a}$. 2

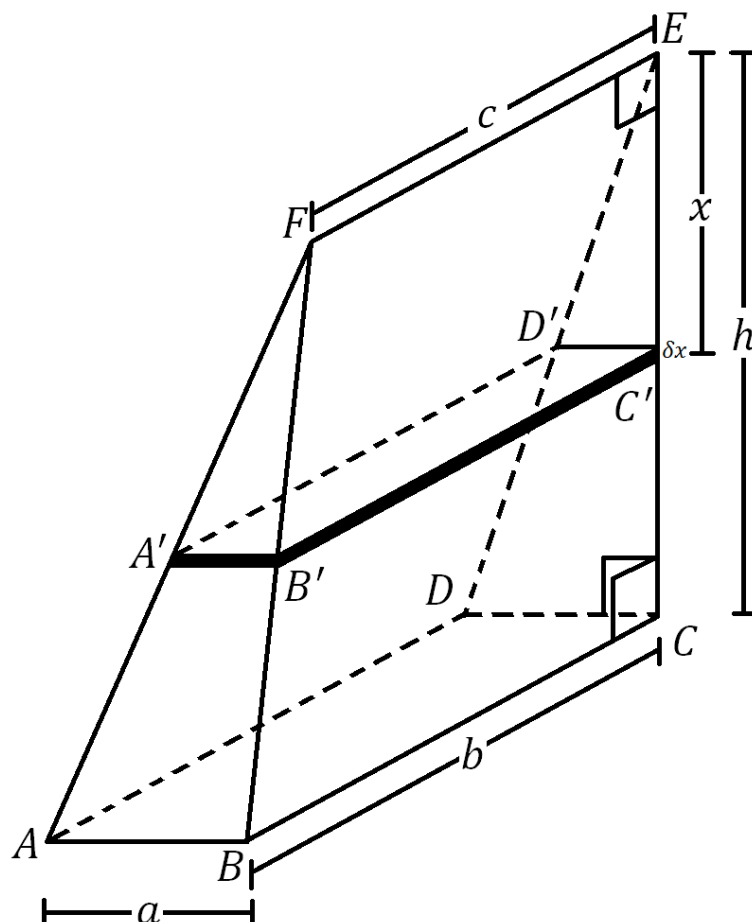
- (f) Given $1, \omega$ and ω^2 are the cube roots of unity and each are represented by the points A_1, A_2 and A_3 respectively on an Argand diagram.

Find the value of $A_1A_2 \times A_1A_3$, where ' A_1A_2 ' represents the length A_1A_2 . 2

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) Consider solid $ABCDEF$ whose height is h , and whose base is a rectangle $ABCD$, where $AB = a$, $BC = b$ and the top edge $EF = c$.



Consider a rectangular slice $A'B'C'D'$ (parallel to the base $ABCD$) which is x units from the top edge with width δx

(Note: $B'C' \parallel BC$ and $A'B' \parallel AB$)

- (i) Show that the volume δV of the slice is given by

$$\delta V = \left(\frac{a}{h}x\right) \left(c + \frac{b-c}{h}x\right) \delta x \quad 3$$

- (ii) Hence show that the volume of the solid $ABCDEF$ is

$$\frac{ha}{6}(2b + c) \quad 2$$

Question 13 continues on the next page

- (b) The acceleration of a motor car on a straight road is $a - bv^2$ where v is the velocity, a and b are positive constants. Let x be the displacement of the motor car from its starting point at time t . Initially, $x = 0, v = 0$.

- (i) Show that at time t , the velocity is given by

$$v = \sqrt{\frac{a}{b}}(1 - e^{-2bx})^{1/2} \quad 3$$

- (ii) Show that the velocity of the motor car has a limiting value of V where V is $\sqrt{\frac{a}{b}}$. 1

- (iii) The velocity p is attained in a displacement l after starting and the velocity q is attained after a further displacement of l where p and q are positive constants and $0 < q < \sqrt{2}p$. Show that

$$V = \frac{p^2}{\sqrt{2p^2 - q^2}} \quad 3$$

- (c) The region bounded by $y = 0, y = e^x, x = 0$ and $x = 2$ is revolved about the line $y = 0$. Find the volume of the resulting solid by using the **cylindrical shell method**. 3

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) Given that $y = \cos^n x \sin nx$
(i) Show that

$$\frac{dy}{dx} = 2n \cos^n x \cos nx - n \cos^{n-1} x \cos(n-1)x \quad 3$$

Hence show that

$$2n \int \cos^n x \cos nx \, dx - n \int \cos^{n-1} x \cos(n-1)x \, dx = \cos^n x \sin nx + C$$

- (ii) Using the result of (i), show that

$$\int_0^{\frac{\pi}{2}} \cos^n x \cos nx \, dx = \frac{1}{2} \int_0^{\frac{\pi}{2}} \cos^{n-1} x \cos(n-1)x \, dx \quad 2$$

- (iii) Let

$$I_n = \int_0^{\frac{\pi}{2}} \cos^n x \cos nx \, dx \quad 2$$

Find I_1 , hence find I_8 .

- (iv) Hence find

$$\int_0^{\frac{\pi}{2}} \sin^8 x \cos 8x \, dx \quad 2$$

- (b) There are eleven men waiting for their turn in a barber shop. Three particular men are A, B and C. There is a row of 11 seats for the customers. Find the number of ways of arranging them so that no two of A, B and C are adjacent. 3

- (c) The curve C has equation $y = \frac{(x-1)^2}{x+2}$

- (i) Obtain the equations of the asymptotes of the curve C. 1

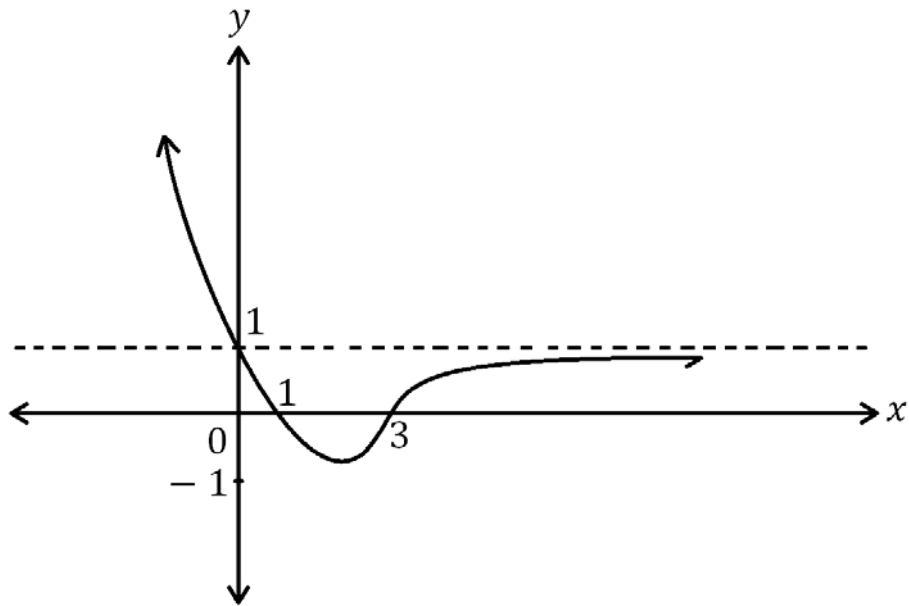
- (ii) On the same diagram, draw a sketch of C and of the curve with equation $y = -\frac{1}{x}$. 2

Deduce the number of real roots of the equation $x^3 - 2x^2 + 2x + 2 = 0$.

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

(a) The graph of $y = f(x)$ is shown below. On separate axes, neatly sketch:



(i) $y = [f(x)]^2$ 1

(ii) $y = f(1 - x)$ 1

(iii) $y = \ln[f(x)]$ 1

(b)

(i) Prove
$$\frac{\cos \theta + i \sin \theta - 1}{\cos \theta + i \sin \theta + 1} = i \tan \frac{\theta}{2}$$
 2

(ii) Find the five roots of the equation $\omega^5 = 1$ and express your answers in the form of $r(\cos \theta + i \sin \theta)$, where $r > 0$ and $-\pi < \theta \leq \pi$. 2

(iii) Hence show that the roots of the equation

$$\left(\frac{2+z}{2-z}\right)^5 = 1 \quad (*)$$
 2

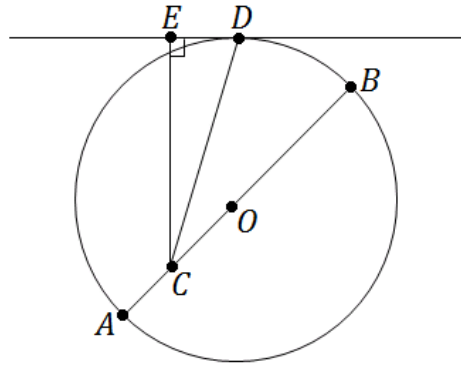
are $2i \tan\left(\frac{k\pi}{5}\right)$, where $k = 0, \pm 1, \pm 2$.

(iv) By expressing the equation in part (iii) (*) in the form of $z^5 + mz^3 + nz = 0$, show that

$$\tan \frac{\pi}{5} \tan \frac{2\pi}{5} = \sqrt{5}$$
 2

Question 15 continues on the next page

(c)



AB is a diameter of the circle and O is the centre. C is a point on AB and D is a point on the circle. DE is the tangent of the circle at D and $CE \perp DE$. Copy the diagram in your booklet.

Extend DC to intersect the circle at F such that DF is a chord of the circle. Similarly, join DO and extend DO to intersect the circle at G . DG now is another diameter for the circle $BDFG$.

- (i) Prove $\triangle CED$ is similar to $\triangle DFG$. 2
- (ii) Hence or otherwise, prove that $AB \times CE = AC \times CB + CD^2$. 2

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

(a) It is given that

$$\frac{r+1}{r-1} = 1 + \frac{2}{r} + \frac{2}{r^2} + \cdots + \frac{2}{r^n} + \frac{2}{r^n(r-1)}$$

(i) Show that

$$\sum_{r=2}^n [\ln(r+1) - \ln(r-1)] = \ln \frac{n(n+1)}{2} \quad 2$$

(ii) Hence prove by mathematical induction that

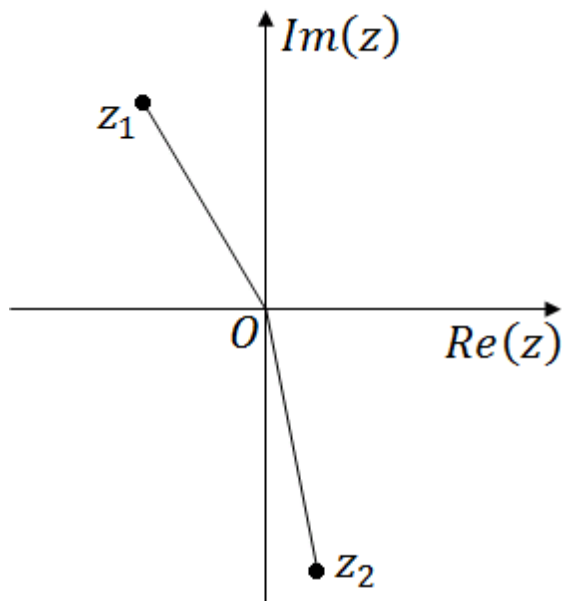
$$\sum_{r=2}^n \ln \left[1 + \frac{2}{r} + \frac{2}{r^2} + \cdots + \frac{2}{r^n} + \frac{2}{r^n(r-1)} \right] = \ln \frac{n(n+1)}{2} \quad \text{for } n = 2, 3, 4, \dots \quad 3$$

Question 16 continues on the next page

- (b) Given that z_1, z_2 and z_3 are three complex numbers which satisfy

$$z_1 + z_2 + z_3 = 0$$

- (i) z_1 and z_2 are indicated in the Argand diagram as shown in the figure below.



Copy the diagram in your booklet and sketch on the same diagram a possible location for z_3 . Explain your decision.

1

- (ii) Given that the arguments of z_1, z_2 and z_3 are α, β and γ respectively, and their moduli are 1, k and $2 - k$ respectively, where $0 < k < 2$. Express z_1, z_2 and z_3 in mod-arg form.

2

- (iii) Prove that

$$\begin{cases} \cos \alpha + k \cos \beta + (2 - k) \cos \gamma = 0 \\ \sin \alpha + k \sin \beta + (2 - k) \sin \gamma = 0 \end{cases}$$

1

- (iv) From (iii), by eliminating α or otherwise, prove that

$$k^2 + (2 - k)^2 + 2k(2 - k) \cos(\beta - \gamma) = 1$$

3

- (v) By considering $|\cos(\beta - \gamma)| \leq 1$, find the range of values of k .

3

End of paper

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

YEAR 12 - EXTENSION 2 - MATHEMATICS - 2013 TRIAL

ANSWER SHEET

INSTRUCTIONS:

- Cross the box that indicates the correct answer

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D

Question (11)

$$\begin{aligned}
 (a) \quad & \int x \tan^{-1} x \, dx \\
 &= \int \tan^{-1} x \frac{d}{dx} \left(\frac{1}{2} x^2 \right) dx \\
 &= \frac{1}{2} x^2 \tan^{-1} x - \frac{1}{2} \int \frac{x^2 dx}{1+x^2} \\
 &= \frac{x^2 \tan^{-1} x}{2} - \frac{1}{2} \int \frac{(1+x^2) - 1}{1+x^2} dx \\
 &= \frac{x^2 \tan^{-1} x}{2} - \frac{1}{2} [x - \tan^{-1} x] \\
 &= \frac{1}{2} (x^2 \tan^{-1} x + \tan^{-1} x - x) + C
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad & \int_{-1/2}^{1/2} \frac{dx}{\sqrt{15-4x-4x^2}} \\
 &= \int_{-1/2}^{1/2} \frac{dx}{\sqrt{-4(x^2+x-\frac{15}{4})}} \\
 &= \int_{-1/2}^{1/2} \frac{dx}{\sqrt{-4(x+\frac{1}{2})^2+4^2}} \\
 &= \frac{1}{2} \int_{-1/2}^{1/2} \frac{dx}{\sqrt{4-(x+\frac{1}{2})^2}}
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad &= \frac{1}{2} \left[\sin^{-1} \left(\frac{x+\frac{1}{2}}{2} \right) \right]_{-1/2}^{1/2} \\
 &= \frac{1}{2} \sin^{-1} \left(\frac{1}{2} \right) = \frac{\pi}{12}
 \end{aligned}$$

$$\begin{aligned}
 (c) \quad & \int_{\pi/2}^{2\pi/3} \frac{d\theta}{2\cos\frac{\theta}{2} - \sin\theta} \\
 & \boxed{\begin{aligned} \text{Let } t &= \tan \frac{\theta}{2} \\ d\theta &= \frac{2dt}{1+t^2}, \\ -\pi < \theta < \pi \end{aligned}}
 \end{aligned}$$

$$\begin{aligned}
 &= \int_1^{\sqrt{3}} \left(\frac{2dt}{1+t^2} \right) \times \left(\frac{2}{t} - \frac{2t}{1+t^2} \right) \\
 &= \int_1^{\sqrt{3}} \frac{dt}{1+t^2} \times \frac{t(1+t^2)}{t+2t^2-2t^2} = \int_1^{\sqrt{3}} t \, dt \\
 &= \left[\frac{1}{2} t^2 \right]_1^{\sqrt{3}} = 1
 \end{aligned}$$

$$\begin{aligned}
 (d) \quad & 3x^2 - x + 8 = A(x^2+1) + (Bx+C)(1-x) \\
 & \text{When } x=1, 2A=10 \Rightarrow A=5 \\
 & A-B=3 \therefore B=2, A+C=8 \Rightarrow C=3 \\
 \therefore \int \frac{(3x^2-x+8)dx}{(1-x)(x^2+1)} &= 5 \int \frac{dx}{1-x} + \int \frac{(2x+3)dx}{x^2+1} \\
 &= -5 \ln(1-x) + \ln(x^2+1) + 3 \tan^{-1} x + C
 \end{aligned}$$

Question (11)

$$(e) \quad \frac{x^2}{4} - y^2 = 1, \quad P(2\sec\theta, \tan\theta), \quad a=2, \quad b=1.$$

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{\sec^2\theta}{2\sec\theta \tan\theta}$$

$\therefore$ Equation of tangent at P

$$y - \tan\theta = \frac{\sec\theta}{2\tan\theta} (x - 2\sec\theta)$$

i.e.

$$(\sec\theta)x - (2\tan\theta)y = 2(\sec^2\theta - \tan^2\theta).$$

$$\left(\frac{\sec\theta}{2}\right)x - (\tan\theta)y = 1$$

The asymptotes are $\begin{cases} y_1 = \frac{1}{2}x_1 \\ y_2 = -\frac{1}{2}x_2 \end{cases}$

(i) When $y_1 = \frac{x_1}{2}$

$$(\sec\theta)x_1 - (\tan\theta)x_1 = 2$$

$$\therefore x_1(\sec\theta - \tan\theta) = 2.$$

$$\therefore x_1 = \frac{2}{\sec\theta - \tan\theta}, \quad y_1 = \frac{1}{\sec\theta - \tan\theta}$$

(ii) When $y_2 = -\frac{x_2}{2}$

$$(\sec\theta)x_2 + (\tan\theta)x_2 = 2$$

$$\therefore x_2 = \frac{2}{\sec\theta + \tan\theta}$$

$$y_2 = \frac{1}{\sec\theta + \tan\theta}$$

$$\begin{aligned} \text{Now, } \frac{x_1 + x_2}{2} &= \frac{1}{\sec\theta - \tan\theta} + \frac{1}{\sec\theta + \tan\theta} \\ &= \frac{2\sec\theta}{\sec^2\theta - \tan^2\theta} \\ &= 2\sec\theta \end{aligned}$$

$$\begin{aligned} \frac{y_1 + y_2}{2} &= \frac{(\sec\theta + \tan\theta) - (\sec\theta - \tan\theta)}{2(\sec^2\theta - \tan^2\theta)} \\ &= \tan\theta. \end{aligned}$$

$\therefore$ The mid-pt of QR is $(2\sec\theta, \tan\theta)$

i.e. the mid-pt is P

Question (11)

$$(f) (a+ib)^2 = 21 - 20i$$

$$a^2 + i2ab - b^2 = 21 - 20i$$

Equate real and imaginary parts

$$\begin{cases} a^2 - b^2 = 21 \\ 2ab = -20 \end{cases}$$

$$\begin{aligned} \text{Now, } (a^2 + b^2)^2 &= (a^2 - b^2)^2 + 4a^2b^2 \\ &= 21^2 + 20^2 \\ &= 841 \end{aligned}$$

$$\therefore a^2 + b^2 = 29$$

$$a^2 - b^2 = 21$$

$$\therefore 2a^2 = 50, \quad a^2 = 25$$

$$\therefore a = \pm 5.$$

$$\text{If } \begin{cases} a = 5, & b = -2 \\ a = -5, & b = 2. \end{cases}$$

Question (12)

(a) $2x^3 + 3x^2 - 5x + 8 = 0$

Let $y = \frac{1}{x}$, $x = \frac{1}{y}$.

$\therefore \frac{2}{y^3} + \frac{3}{y^2} - \frac{5}{y} + 8 = 0$

$\Rightarrow 8y^3 - 5y^2 + 3y + 2 = 0$

(b) $z = 3\cos\theta + j3\sin\theta$

$1-z = (1-3\cos\theta) - j3\sin\theta$

$\overline{1-z} = (1-3\cos\theta) + j3\sin\theta$

$$\frac{1}{1-z} = \frac{1}{(1-3\cos\theta) - j3\sin\theta} \times \frac{(1-3\cos\theta) + j3\sin\theta}{(1-3\cos\theta) + j3\sin\theta}$$

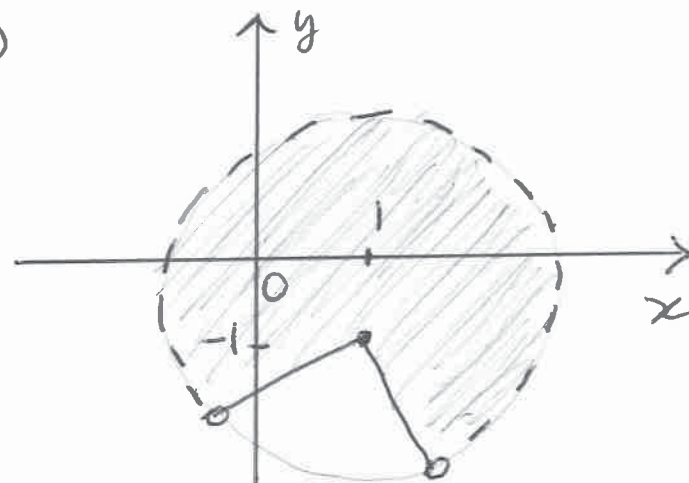
$$= \frac{1-3\cos\theta + j3\sin\theta}{(1-3\cos\theta)^2 + 9\sin^2\theta}$$

$$= \frac{(1-3\cos\theta) + j3\sin\theta}{1-6\cos\theta + 9\cos^2\theta + 9\sin^2\theta}$$

$$= \frac{1-3\cos\theta + j3\sin\theta}{10-6\cos\theta}$$

$\therefore \operatorname{Im}\left(\frac{1}{1-z}\right) = \frac{3\sin\theta}{2(5-3\cos\theta)}$

(c)



(d) $z_1 = 2j$, $z_2 = 1+3j$

(i) $k=1$, $z = z_1 + (z_2 - z_1) = z_2$

$\therefore$ locus of z is a point $(1, 3)$.

(ii) $0 < k < 1$, locus of z is the line interval between $(0, 2)$ and $(1, 3)$

(iii) $k \in \mathbb{R}$, locus of z is the line through $(0, 2)$ and $(1, 3)$.
i.e. $y = x + 2$

Question (12)

$$(e) (i) \sum \alpha_i = 0 \Rightarrow \alpha + \beta + 2\alpha - 2\beta = 0$$

$$\therefore 3\alpha = \beta$$

$$\sum_{1 \leq i < j} \alpha_i \alpha_j = a, \therefore \alpha\beta + 2\alpha(\alpha - \beta) + 2\beta(\alpha - \beta) = a.$$

Expanding,

$$\alpha\beta + 2\alpha^2 - 2\alpha\beta + 2\alpha\beta - 2\beta^2 = a.$$

but $3\alpha = \beta$ from (i),

$$\therefore 3\alpha^2 + 2\alpha^2 - 18\alpha^2 = a$$

$$\Rightarrow a = -13\alpha^2$$

Product of roots = -b

$$\therefore 2\alpha\beta(\alpha - \beta) = -b.$$

$$\text{but } 3\alpha = \beta \therefore 6\alpha^2(-2\alpha) = -b$$

$$\Rightarrow b = 12\alpha^3$$

$$\therefore \frac{b}{a} = \frac{-12\alpha}{-13} \Rightarrow \alpha = \frac{-13b}{12a}.$$

$$\text{but } 3\alpha = \beta \therefore \beta = 3\left(\frac{-13b}{12a}\right)$$

$$= \frac{-13b}{4a}.$$

$$(ii) (\text{cont.}) \text{ Now, } 2(\alpha - \beta)$$

$$= 2\left(\frac{-13b}{12a} + \frac{13b}{4a}\right)$$

$$= \frac{13b}{3a}$$

$\therefore$ roots of $P(x)$

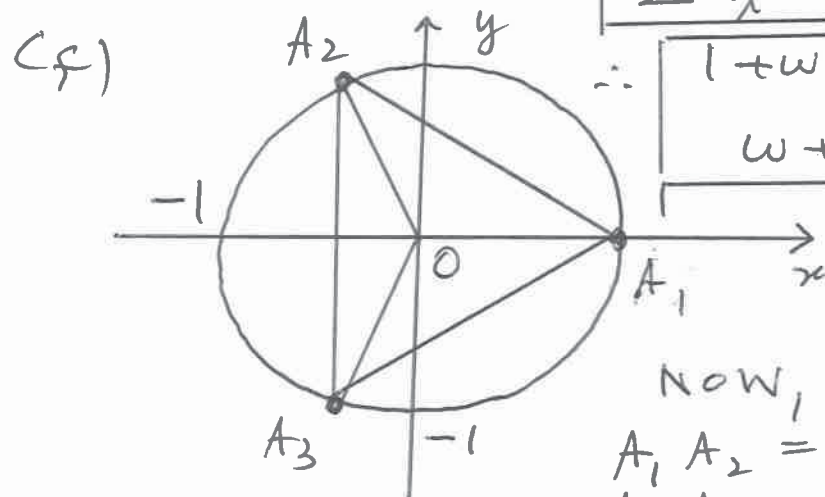
$$\text{are: } \frac{-13b}{a}, \frac{-13b}{4a}, \frac{13b}{3a}$$

$$z^3 - 1 = 0$$

$$\sum \alpha_i = 0$$

$$\therefore 1 + w + w^2 = 0$$

$$w + w^2 = -1$$



Now,

$$A_1 A_2 = |w - 1|$$

$$A_1 A_3 = |w^2 - 1|$$

$$\therefore A_1 A_2 \times A_1 A_3 = |w - 1| |w^2 - 1|$$

$$= |w^3 - w - w^2 + 1|$$

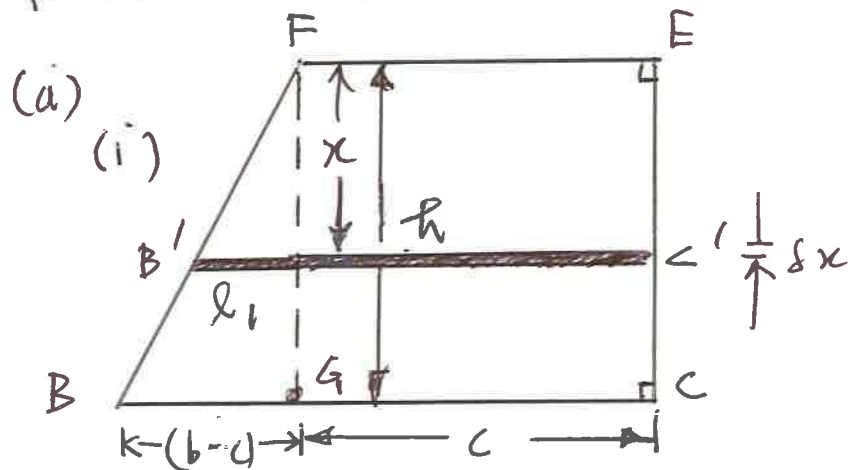
$$= |2 - (w + w^2)|$$

$$= |2 - (-1)|$$

$$= |3| = 3.$$

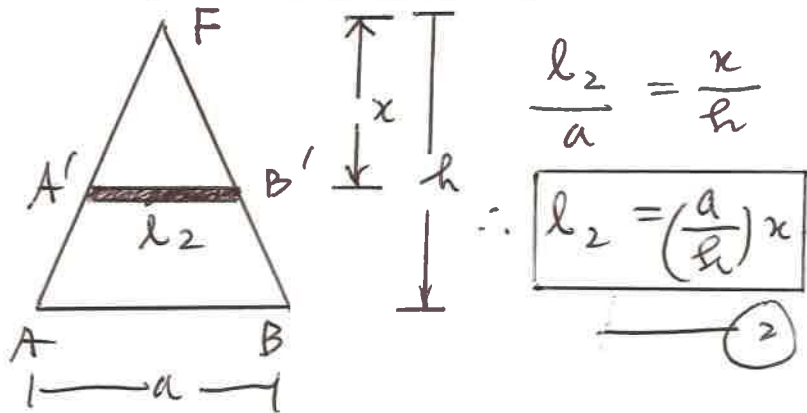
$$\therefore w + w^2 = -1$$

Question (13)



$$\text{In } \triangle FBG, \frac{l_1}{b-c} = \frac{x}{h}$$

$$\therefore l_1 = \left(\frac{b-c}{h} \right) x \quad \text{--- (1)}$$



$$\delta V = (A'B')(B'C') \delta x \quad \text{--- (3)}$$

$$A'B' = l_2 = \left(\frac{a}{h} \right) x$$

$$B'C' = c + \left(\frac{b-c}{h} \right) x$$

$$\therefore \delta V = \left(\frac{a}{h} x \right) \left[c + \left(\frac{b-c}{h} \right) x \right] \delta x \quad \text{--- (4)}$$

$$\text{ii) } V = \lim_{\delta x \rightarrow 0} \sum_0^h \left(\frac{a}{h} x \right) \left(c + \left(\frac{b-c}{h} \right) x \right) \delta x$$

$$\therefore V = \int_0^h \left\{ \left(\frac{ac}{h} \right) x + \left[\frac{a(b-c)}{h^2} \right] x^2 \right\} dx$$

$$= \left[\left(\frac{ac}{h} \right) \frac{x^2}{2} + \frac{a(b-c)}{h^2} \frac{x^3}{3} \right]_0^h$$

$$= \frac{3ach + 2ah(b-c)}{6}$$

$$= \frac{ha}{6} (2b+c) \quad \text{--- (5)}$$

Question (13)

(b) $v \frac{dv}{dx} = a - bv^2$.
Separating variables, we have

$$\int \frac{v}{a - bv^2} dv = \int dx.$$

$$\text{i.e.} \int \frac{1}{a - bv^2} d\left(\frac{a - bv^2}{-2b}\right) = \int dx.$$

$$\therefore -\frac{1}{2b} \ln(a - bv^2) = x + c$$

When $x=0$, $x=0$ and $v=0$,
we have $c = -\frac{1}{2b} \ln a$.

$$\therefore -\frac{1}{2b} \ln(a - bv^2) = x - \frac{1}{2b} \ln a$$

$$\text{i.e.} \boxed{v = \sqrt{\frac{a}{b}} (1 - e^{-2bx})^{\frac{1}{2}}} \quad (\because v > 0).$$

(ii) As $x \rightarrow \infty$ $v \rightarrow \sqrt{\frac{a}{b}}$.

Since $e^{-2bx} \rightarrow 0$
The limiting value v of the
motor car is $\sqrt{\frac{a}{b}}$.

(iii) When $v = \sqrt{\frac{a}{b}}$

$$v = \sqrt{\frac{a}{b}} \sqrt{1 - e^{-2bx}}$$

When $v=p$ and $x=l$

$$\therefore p = \sqrt{\frac{a}{b}} \sqrt{1 - e^{-2bl}}$$

$$\text{i.e.} e^{-2bl} = \frac{v^2 - p^2}{v^2}$$

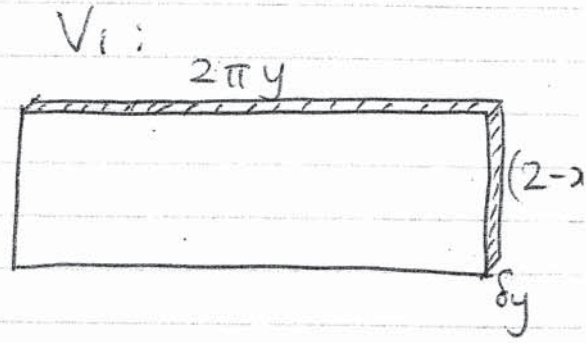
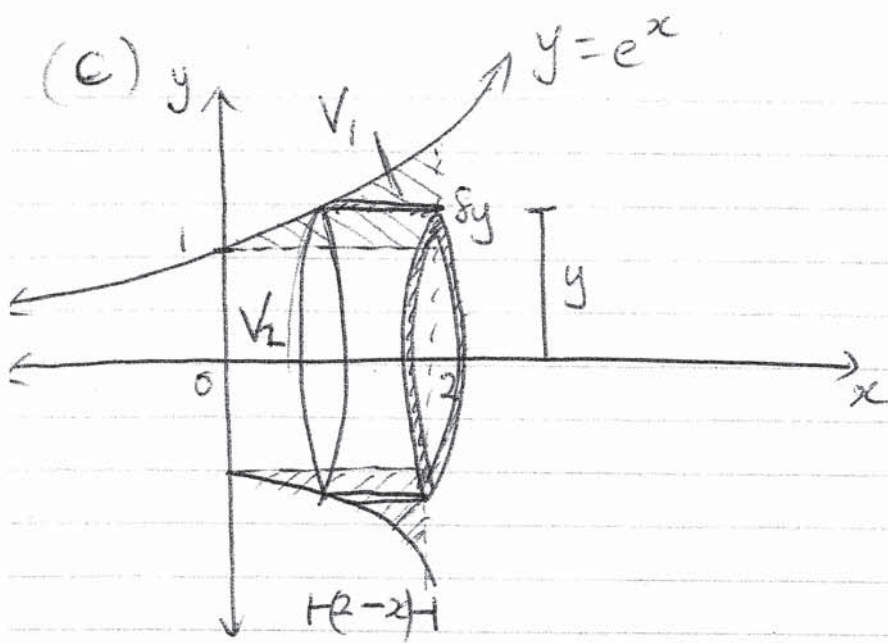
When $v=q$ and $x=l+l=2l$,
we have $q = \sqrt{\frac{a}{b}} \sqrt{1 - e^{-4bl}}$

$$\text{i.e.} e^{-4bl} = \frac{v^2 - q^2}{v^2}$$

Eliminating displacement l ,
we have $\left(\frac{v^2 - p^2}{v^2}\right)^2 = \frac{v^2 - q^2}{v^2}$

$$\text{i.e.} v = \frac{p^2}{\sqrt{2p^2 - q^2}} \quad (\because v > 0)$$

(4)



$$V_1 = \int_1^{e^2} 2\pi y (2-x) \cdot dy$$

$$= 2\pi \int_1^{e^2} y(2 - \ln y) \cdot dy$$

$$= 2\pi \int_1^{e^2} 2y - y \ln y \cdot dy$$

$$= 2\pi \left\{ [y^2]_1^{e^2} - \int_1^{e^2} y \ln y \cdot dy \right\}$$

$$= 2\pi \left\{ e^4 - 1 - \left(\left[\frac{y^2 \ln y}{2} \right]_1^{e^2} - \int_1^{e^2} \frac{y^2}{2} \cdot \frac{1}{y} \cdot dy \right) \right\} \quad (1)$$

$$= 2\pi \left\{ e^4 - 1 - \left(e^4 - \left[\frac{y^2}{4} \right]_1^{e^2} \right) \right\}$$

$$= 2\pi \left\{ e^4 - 1 - \left(e^4 - \frac{e^4}{4} + \frac{1}{4} \right) \right\}$$

$$= 2\pi \left(\frac{e^4}{4} - \frac{5}{4} \right) \quad (1)$$

$$= \frac{\pi}{2} (e^4 - 5)$$

$$V_2 = \pi (1)^2 (2)$$

$$V_2 = 2\pi$$

$$V_{\text{Total}} = V_1 + V_2$$

$$= \frac{\pi}{2} (e^4 - 5) + 2\pi$$

$$= \boxed{\frac{\pi}{2} (e^4 - 1)} \quad (1)$$

Question (14)

(a) $y = \cos^n x \sin nx$.

(i) $\frac{dy}{dx} = n \cos^n x \cos nx - n \sin nx \cos^{n-1} x \sin x$

$$= 2n \cos^n x \cos nx - n \cos^{n-1} x (\sin nx \sin x + \cos nx \cos nx)$$

$$= 2n \cos^n x \cos nx - n \cos^{n-1} x \cos(n-1)x$$

$$\int d(\cos^n x \sin nx)$$

$$= \int 2n \cos^n x \cos nx dx - n \int \cos^{n-1} x \cos(n-1)x dx$$

$$= 2n \int \cos^n x \cos nx dx - n \int \cos^{n-1} x \cos(n-1)x dx$$

$$= \cos^n x \sin nx + C$$

(ii) $\left[\cos^n x \sin nx \right]_0^{\pi/2}$

$$= 2n \int_0^{\pi/2} \cos^n x \cos nx dx - n \int_0^{\pi/2} \cos^{n-1} x \cos(n-1)x dx$$

$$\therefore 0 = 2n \int_0^{\pi/2} \cos^n x \cos nx dx - n \int_0^{\pi/2} \cos^{n-1} x \cos(n-1)x dx$$

$$\therefore \int_0^{\pi/2} \cos^n x \cos nx dx = \frac{1}{2} \int_0^{\pi/2} \cos^{n-1} x \cos(n-1)x dx$$

(iii) $I_n = \int_0^{\pi/2} \cos^n x \cos nx dx$

$$\therefore I_1 = \int_0^{\pi/2} \cos^2 x dx = \int_0^{\pi/2} \left(\frac{1 + \cos 2x}{2} \right) dx$$

$$\therefore I_1 = \left[\frac{x}{2} + \frac{\sin 2x}{4} \right]_0^{\pi/2}$$

$$= \frac{\pi}{4}$$

From (ii) $I_8 = \frac{1}{2} I_7$

$$= \frac{1}{2^2} I_6$$

$$= \frac{1}{2^3} I_5$$

$$= \frac{\pi}{512}$$

(iv) Let $y = \frac{\pi}{2} - x$, $dy = -dx$

and $x = \frac{\pi}{2} - y$. When $x=0$, $y = \frac{\pi}{2}$;
When $x = \frac{\pi}{2}$, $y = 0$.

$$\therefore \int_0^{\pi/2} \sin^8 x \cos 8x dx$$

$$= \int_{\pi/2}^0 \sin^8 \left(\frac{\pi}{2} - y \right) \cos \left[8 \left(\frac{\pi}{2} - y \right) \right] (-dy)$$

$$= \int_0^{\pi/2} \cos^8 y \cos 8y dy = \frac{\pi}{512}$$

Question (14)

(b) We first arrange the 8 persons (excluding A, B and C) in a row in $8!$ ways. Fix one of these ways, say

 x_1 x_2 x_3 x_4 x_5 x_6 x_7 x_8

(1) (2) (3) (4) (5) (6) (7) (8) (9)

We now consider A. There are 9 ways to place A in one of 9 boxes, say box (4):

 x_1 x_2 x_3 $\textcircled{A}$ x_4 x_5 x_6 x_7 x_8

(1) (2) (3) (4) (5) (6) (7) (8) (9)

Next, consider B. Since A and B cannot be adjacent, B can be placed only in one of the remaining 8 boxes. Like wise, C can be placed only in one of the remaining 7 boxes.

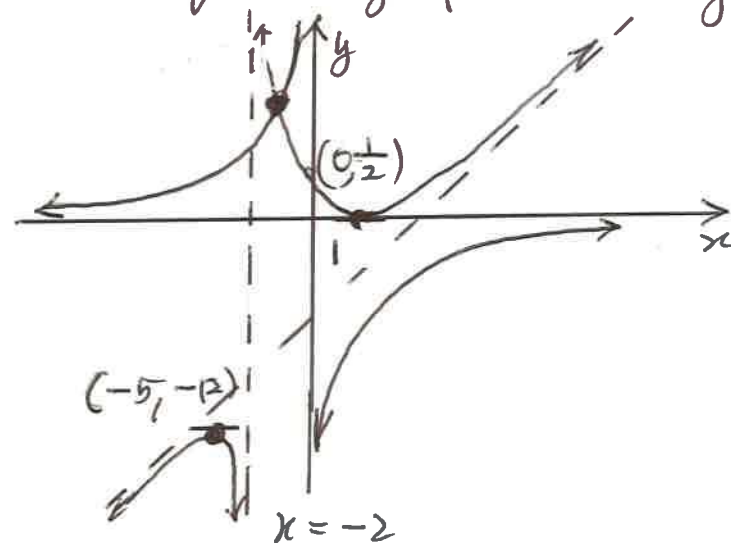
$$\therefore (8!) 9 \times 8 \times 7 = 20321280$$

(c) Since $x \neq -2$, a vertical asymptote is $x = -2$

(i)
$$y = \frac{(x-1)^2}{x+2} = \frac{x^2 - 2x + 1}{x+2} = x - 4 + \frac{9}{x+2}$$

An oblique asymptote is $y = x - 4$

(ii)



Given: $x^3 - 2x^2 + 2x + 2 = 0$

$$\Rightarrow x(x^2 - 2x + 1) + x + 2 = 0$$

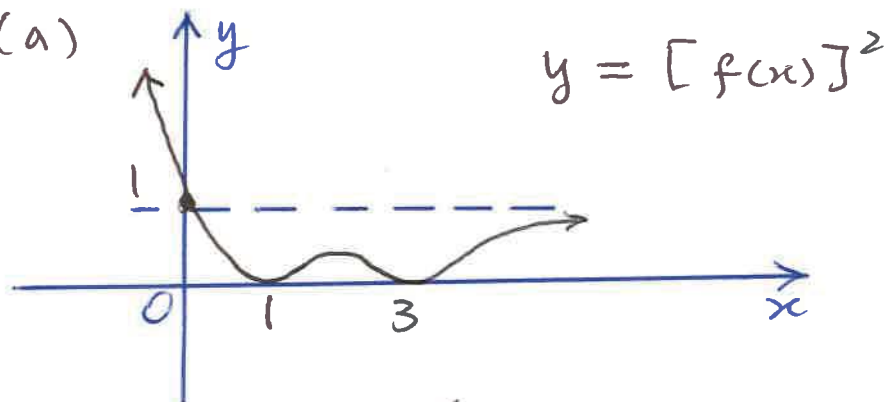
$$x(x^2 - 2x + 1) = -(x + 2)$$

$$\therefore \frac{x^2 - 2x + 1}{x + 2} = -\frac{1}{x}$$

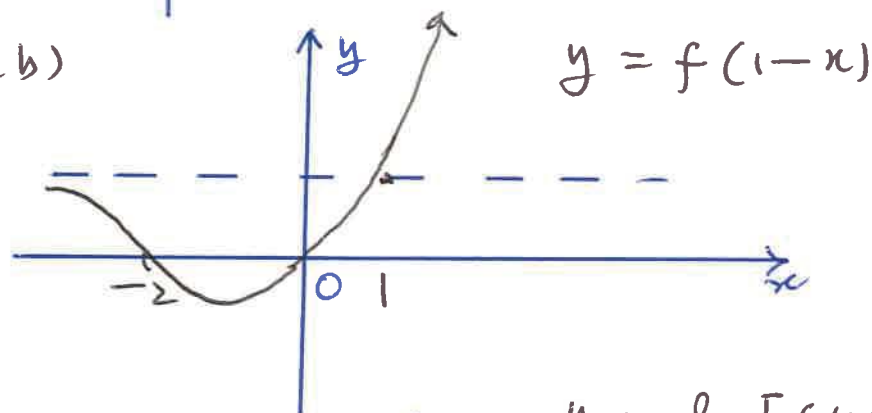
Since there is one intersection between the graph of $\frac{1}{x}$ and $y = -\frac{1}{x}$, there is one real root for $x^3 - 2x^2 + 2x + 2 = 0$

Question (15)

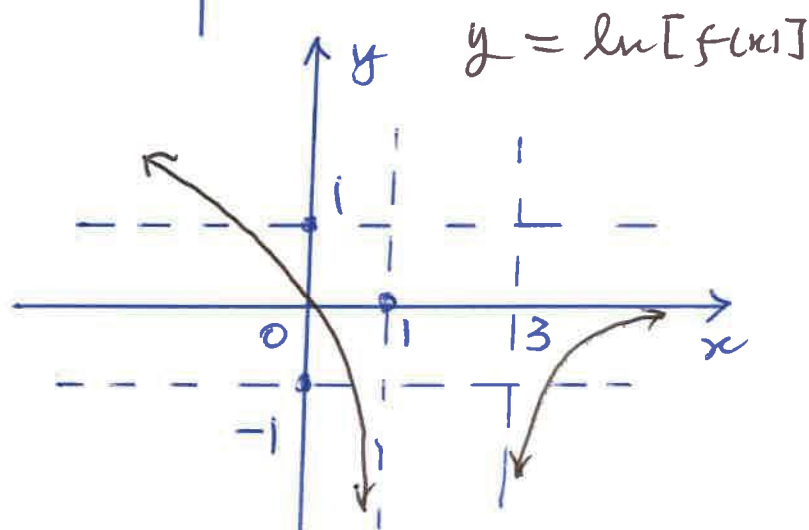
(a)



(b)



(c)



$$(b) \frac{\cos \theta + i \sin \theta - 1}{\cos \theta + i \sin \theta + 1}$$

$$= \frac{(\cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2}) + i(2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}) - (\sin^2 \frac{\theta}{2} + \cos^2 \frac{\theta}{2})}{(\cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2}) + i(2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}) + (\sin^2 \frac{\theta}{2} + \cos^2 \frac{\theta}{2})}$$

$$= \frac{-2 \sin^2 \frac{\theta}{2} + i 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \cos^2 \frac{\theta}{2} + i 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}$$

$$= \frac{i 2 \sin \frac{\theta}{2} (\cancel{i \sin \frac{\theta}{2}} + \cos \frac{\theta}{2})}{2 \cos \frac{\theta}{2} (\cancel{\cos \frac{\theta}{2}} + i \sin \frac{\theta}{2})}$$

$$= i \tan \frac{\theta}{2}$$

(ii) Let $w = r \operatorname{cis} \theta$, $w^5 = 1$

$$(r \operatorname{cis} \theta)^5 = 1, \quad r^5 \operatorname{cis} 5\theta = \operatorname{cis} (2n\pi)$$

$$\therefore r^5 = 1 \Rightarrow r = 1$$

$$\text{Now, } 5\theta = 2n\pi, \quad \theta = \frac{2n\pi}{5} \quad (n=0, 1, 2, 3, 4)$$

$$\therefore \text{roots are } \operatorname{cis} 0, \operatorname{cis} \left(-\frac{2\pi}{5}\right)$$

$$\operatorname{cis} \left(-\frac{4\pi}{5}\right), \operatorname{cis} \left(\frac{2\pi}{5}\right), \operatorname{cis} \left(\frac{4\pi}{5}\right)$$

Question (15)

(iii) Let $\omega = \left(\frac{2+z}{2-z} \right)$

$$2\omega - \omega z = 2 + z$$

$$2\omega - 2 = \omega z + z$$

$$2(\omega - 1) = z(\omega + 1)$$

$$\therefore z = 2 \left(\frac{\omega - 1}{\omega + 1} \right)$$

Now, from (ii) $\omega = \text{cis} \left(\frac{2n\pi}{5} \right)$
for $n = \pm 1, \pm 2, 0$.

From (ii) we have

$$\omega = \cos \frac{2k\pi}{5} + i \sin \frac{2k\pi}{5},$$

$k = 0, \pm 1, \pm 2$. When $k=0$, $z=0$

When $k=1$, $z = 2 \left[\frac{\cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} - 1}{\cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} + 1} \right]$
 $= 2i \tan \frac{2\pi}{5}$

When $k=-4$, $z = 2 \left[\frac{\cos \left(-\frac{4\pi}{5} \right) + i \sin \left(-\frac{4\pi}{5} \right) - 1}{\cos \left(-\frac{4\pi}{5} \right) + i \sin \left(-\frac{4\pi}{5} \right) + 1} \right]$

$$\therefore z = 2i \tan \left(\frac{n\pi}{5} \right) \text{ for } n = 0, \pm 1, \pm 2.$$

$$(iv) \left(\frac{2+z}{2-z} \right)^5 = 1$$

$$(2+z)^5 = (2-z)^5$$

$$32 + 80z + 80z^2 + 40z^3 + 10z^4 + z^5 = 32 - 80z + 80z^2 - 40z^3 + 10z^4 - z^5$$

$$\therefore 2z^5 - 80z^3 + 160z = 0$$

$$z^5 - 40z^3 + 80z = 0$$

$$\therefore z(z^4 - 40z^2 + 80) = 0$$

[Note: $M = -40$, $n = 80$].

Product of roots (excluding $z=0$) = 80

i.e

$$2^4 (\lambda^4) \tan \left(\frac{2\pi}{5} \right) \tan \left(-\frac{2\pi}{5} \right) \times \tan \left(\frac{\pi}{5} \right) \tan \left(-\frac{\pi}{5} \right) = 80$$

$$\therefore 16 \left[-\tan^2 \frac{2\pi}{5} \right] \left[-\tan^2 \frac{\pi}{5} \right] = 80$$

$$\therefore \tan^2 \left(\frac{2\pi}{5} \right) \tan^2 \left(\frac{\pi}{5} \right) = 5$$

$$\text{i.e } \tan \left(\frac{\pi}{5} \right) \tan \left(\frac{2\pi}{5} \right) = \sqrt{5}$$

Question 15(c)

Produce DC to intersect the circle at F.

Similarly, extend DO to intersect the circle at G.

$$\text{Now } AC \cdot CB = FC \cdot CD.$$

(product of intercepts of intersecting chords are equal)

Now

$$FC \cdot CD + CD^2$$

$$= CD (FC + CD)$$

$$= CD \cdot FD.$$

$$AB \times CE$$

$$= 2(\text{radius}) \times CE$$

$$= DG \cdot CE.$$

To prove.

$$AB \times CE = AC \times CB + CD^2.$$

it is sufficient to prove

$$CD \cdot FD = CE \cdot DG$$

$$\text{or } \frac{FD}{CE} = \frac{DG}{CD}.$$

Now In $\triangle CED$ and $\triangle DFG$.

$$\angle DFG = 90^\circ \text{ (Angle in a semi-circle)}$$

$$= \angle CED$$

$$\angle EDC = \angle DGF \text{ (alternate segment theorem.)}$$

$$\therefore \triangle CED \parallel \triangle DFG.$$

(equiangular).

$$\therefore \frac{FD}{CE} = \frac{DG}{CD}.$$

(corresponding sides of similar triangles $\triangle CED$ and $\triangle DFG$ are in the same ratios.)

$$\therefore AC \cdot CD + CD^2 = AB \cdot CE.$$

Question (16) (a)

$$\begin{aligned} \text{(i)} \quad & \sum_{r=2}^n [\ln(r+1) - \ln(r-1)] \\ &= (\ln 3 - \ln 1) + (\ln 4 - \ln 2) \\ &+ (\ln 5 - \ln 3) + (\ln 6 - \ln 4) \\ &+ \dots + [\ln(n-1) - \ln(n-3)] \\ &+ [\ln(n) - \ln(n-2)] \\ &+ [\ln(n+1) - \ln(n-1)] \\ &= -\cancel{\ln 1} - \ln 2 + \ln(n) + \ln(n+1) \\ &= \frac{\ln n(n+1)}{2} \end{aligned}$$

(ii) $\therefore$

$$\frac{r+1}{r-1} = 1 + \frac{2}{r} + \dots + \frac{2}{r^n} + \frac{2}{r^n(r-1)}$$

(given)

$$\begin{aligned} \therefore \sum_{r=2}^n \ln \left[1 + \frac{2}{r} + \dots + \frac{2}{r^n} + \frac{2}{r^n(r-1)} \right] \\ = \sum_{r=2}^n \ln \left(\frac{r+1}{r-1} \right) \end{aligned}$$

Let $P(n)$ be the proposition

$$\text{that } \sum_{r=2}^n \ln \left(\frac{r+1}{r-1} \right) = \ln \frac{n(n+1)}{2}$$

When $n=2$, L.H.S. = $\ln 3$ = R.H.S.

$\therefore P(2)$ is true

Assume $P(k)$ is true

$$\text{i.e. } \sum_{r=2}^k \ln \left(\frac{r+1}{r-1} \right) = \ln \frac{k(k+1)}{2}$$

Prove true for $n=k+1$.

Now when $n=k+1$, we have

$$\begin{aligned} \sum_{r=2}^{k+1} \ln \left(\frac{r+1}{r-1} \right) &= \sum_{r=2}^k \ln \left(\frac{r+1}{r-1} \right) \\ &+ \ln \left(\frac{k+2}{k} \right) \end{aligned}$$

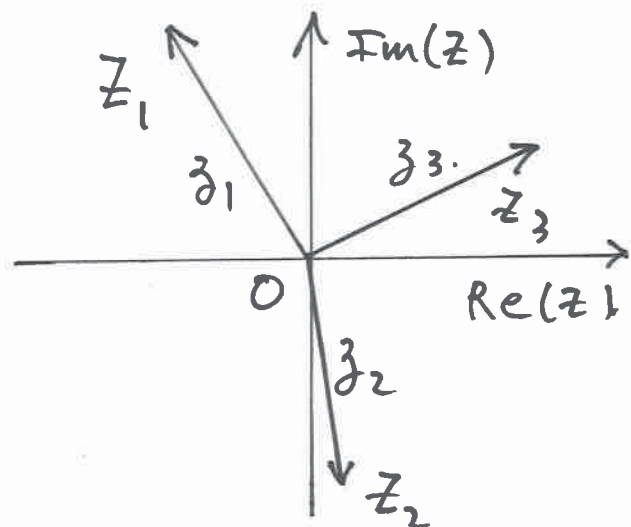
$$= \ln \left[\frac{k(k+1)}{2} \right] + \ln \left(\frac{k+2}{k} \right)$$

$$= \ln \left[\frac{k(k+1)(k+2)}{2k} \right]$$

$$= \frac{\ln \{ (k+1) [(k+1)+1] \}}{2}$$

If the propn. is true for $n=k$, then it is true for $n=k+1$. By the principle of M.I. it is true $\forall n \geq 2$.

Question (16) (b)



(i)

$$z_1 + z_2 + z_3 = 0$$

$$\vec{OZ_1} + \vec{OZ_2} + \vec{OZ_3} = 0$$

(ii)

$$z_1 = \cos \alpha + j \sin \alpha$$

$$z_2 = k \cos \beta + j k \sin \beta$$

$$z_3 = (2-k) \cos \gamma + j (2-k) \sin \gamma$$

where $0 < k < 2$

(iii) $\therefore z_1 + z_2 + z_3 = 0$

$$\therefore \cos \alpha + j \sin \alpha + (k \cos \beta + j k \sin \beta) + ((2-k) \cos \gamma + j (2-k) \sin \gamma) = 0$$

$$\therefore \text{Real (LHS)} = \text{Im (LHS)} = 0$$

So

$$\begin{cases} \cos \alpha + k \cos \beta + (2-k) \cos \gamma = 0 \\ \sin \alpha + k \sin \beta + (2-k) \sin \gamma = 0 \end{cases}$$

Make $\cos \alpha$, $\sin \alpha$ the subject

$$\therefore \cos \alpha = -[k \cos \beta + (2-k) \cos \gamma] \quad \text{--- (1)}$$

$$\sin \alpha = -[k \sin \beta + (2-k) \sin \gamma] \quad \text{--- (2)}$$

(iv) Square (1) & (2) and find its sum

$$\therefore \cos^2 \alpha + \sin^2 \alpha$$

$$= k^2 \cos^2 \beta + (2-k)^2 \cos^2 \gamma + 2k(2-k) \cos \beta \cos \gamma + k^2 \sin^2 \beta + (2-k)^2 \sin^2 \gamma + 2k(2-k) \sin \beta \sin \gamma$$

i.e.

$$1 = k^2 (\sin^2 \beta + \cos^2 \beta) + (2-k)^2 (\sin^2 \gamma + \cos^2 \gamma) + 2k(2-k) [\cos \beta \cos \gamma + \sin \beta \sin \gamma]$$

$$\therefore 1 = k^2 + (2-k)^2 + 2k(2-k) \cos(\beta - \gamma) \quad \text{--- (3)}$$

Expand (3) we have

$$(k^2) + (4 - 4k + k^2) + (4k - 2k^2) \Leftrightarrow (\beta - \gamma) = 1$$

$$\therefore (2k^2 - 4k) [1 - \Leftrightarrow (\beta - \gamma)] = -3.$$

$$\therefore 2k^2 - 4k = \frac{-3}{[1 - \Leftrightarrow (\beta - \gamma)]} \quad \text{--- (4)}$$

$$\text{Let } \boxed{\delta = \frac{-3}{1 - \Leftrightarrow (\beta - \gamma)}} \quad \text{--- (5)}$$

Now $|\Leftrightarrow (\beta - \gamma)| \leq 1$

i.e.

$$-1 \leq \Leftrightarrow (\beta - \gamma) \leq 1 \quad \text{--- (6)}$$

Substitute the extreme values of (6) $\Rightarrow$ the first approximation of (5) becomes

$$-\infty < \delta \leq -\frac{3}{2} \quad \text{--- (7)}$$

Now from (4) we can form a quadratic equation

$$2k^2 - 4k - \delta = 0$$

where δ is given by (5).

$$\therefore k = \frac{4 \pm \sqrt{16 + 8\delta}}{4}$$

$$\text{i.e. } k = \frac{4 \pm \sqrt{1 + \frac{\delta}{2}}}{4} \quad \text{--- (8)}$$

$$k = 1 \pm \sqrt{1 + \frac{\delta}{2}} \quad \text{Solving for } k \in \mathbb{R},$$

$$1 + \frac{\delta}{2} \geq 0$$

$$\delta \geq -2 \quad \text{--- (9)}$$

$\therefore$ A refinement for δ

$$\text{is: } -2 \leq \delta \leq -\frac{3}{2}.$$

So extreme values of δ are $-2, -\frac{3}{2}$.

$$\therefore 1 - \sqrt{\frac{1}{4}} \leq k \leq 1 + \sqrt{\frac{1}{4}}$$

$$\text{i.e. } \boxed{\frac{1}{2} \leq k \leq \frac{3}{2}} \quad \text{--- (10)}$$